Mock Final Exam – Machine Learning (MDS + BDMA + MIRI)

Duration: 1h30m

All questions have equal weight. Justify all your answers. Be concise and clear.

Question 1

You are given a dataset with 500 data points and 50 features. The target variable is binary.

(a) What model(s) would you expect to perform well on this dataset and why?

(b) Would you apply feature scaling? For which models specifically?

(c) What regularization method would you choose for logistic regression and why?

Question 2

Describe how **early stopping** can be viewed as a form of regularization in neural networks. What does it control in terms of the bias-variance tradeoff?

Question 3

Consider a model selection strategy using k-fold cross-validation. Instead of averaging validation errors across folds, you decide to:

(a) Use the **maximum** validation error across folds

(b) Use the **standard deviation** of the errors

Explain what effect each of these has on model selection. In what scenarios might they be preferable over the mean?

Question 4

Construct or describe a 2D dataset (two features) where:

- A decision tree of depth 1 misclassifies some data points
- But a linear SVM with a well-chosen kernel performs perfectly

Sketch or describe the data points and explain why this happens.

Question 5

Mark the following statements as **True** or **False** and briefly justify each:

(a) Ridge regression reduces the number of features used in the model

(b) Logistic regression can model non-linear decision boundaries

(c) Neural networks require input feature scaling to train effectively

(d) Naive Bayes assumes features are conditionally independent given the class

(e) The error on the training set is always smaller than on the test set

(f) Decision trees are sensitive to feature scaling

(g) k-NN has high variance and low bias

(h) More hidden units in an MLP always improve generalization